PROPOSAL FOR SPACE WEEK

MISSION MARS: FROM EARTH TO THE RED PLANET

An Interactive Journey Through Space Science, Exploration and Technology

Proposed by:
Academic Outreach Programme Team
Pondicherry University

In collaboration with:
Pondicherry Science Forum

Proposed Duration: One Day
Proposed Time: 9:30 AM – 4:30 PM
Target Audience: Students of Classes VIII and IX
Expected Participants: Approximately 50 to 100 students
Number of Student Presenters/Facilitators: 7

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1. INTRODUCTION

Space science has the unique ability to inspire students to ask questions about the world around them and the universe beyond it. However, space science is often introduced to students primarily through textbooks, lectures and presentations.

This programme proposes a different approach.

"Mission Mars: From Earth to the Red Planet" is designed as an immersive and activity-oriented space science outreach programme in which students follow a simulated interplanetary mission from its beginning on Earth to exploration of Mars and the return of scientific information to Earth.

Instead of presenting space science as a collection of isolated topics, the programme connects different scientific concepts into one continuous mission.

Students will first understand the formation of Earth and the Solar System, learn how telescopes allow us to observe distant objects, understand why Mars is an important target for exploration, and then follow the major stages of a Mars mission through practical demonstrations, models and interactive activities.

The programme will combine **astronomy, planetary science, aerospace engineering, robotics, communication technology and scientific problem-solving** in a format appropriate for students of Classes VIII and IX.

2. CENTRAL CONCEPT

The entire programme will follow one central question:

> **“Can we successfully send a mission from Earth to Mars, explore the planet and bring the mission safely back to Earth?”**

The students will follow the mission through the following stages:

Earth → Solar System → Astronomy → Mars → Mission Planning → Rocket → Launch → Mars Landing → Rover Exploration → Communication → Return to Earth

Each stage will be handled by a different presenter/facilitator with appropriate demonstrations and models.

The seven presenters will function as different members of a fictional **Mars Mission Team**, while the students will act as mission participants and observers.

3. OBJECTIVES

The programme aims to:

1. Introduce students to the basic concepts of astronomy and space science.
2. Explain the formation of Earth and the Solar System in a simple and engaging manner.
3. Introduce students to planets and their major characteristics.
4. Demonstrate how telescopes are used to observe distant astronomical objects.
5. Explain why Mars is considered an important target for planetary exploration.
6. Introduce the basic principles of rocket propulsion and space travel.
7. Demonstrate the challenges involved in launching and landing spacecraft.
8. Introduce students to planetary rovers and robotic exploration.
9. Demonstrate the importance of communication and navigation in space missions.
10. Encourage scientific curiosity, teamwork, creativity and problem-solving.
11. Introduce students to the different careers and technologies involved in space exploration.
12. Provide students with a memorable learning experience beyond conventional classroom teaching.

4. TARGET AUDIENCE

The programme is primarily designed for:

- Students of Classes VIII and IX
- Approximately 50 to 100 participants
- Students with varying levels of prior knowledge

The programme will therefore assume **no advanced knowledge of astronomy or space science**.

All scientific concepts will be introduced progressively through simple explanations, demonstrations, visual materials and practical activities.

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5. PROGRAMME STRUCTURE

The programme will consist of seven major scientific sessions connected through a single Mars mission storyline.

SESSION 1 — OUR HOME: EARTH AND THE SOLAR SYSTEM

Presenter 1 — Earth & Planetary Science

The programme begins by introducing the students to their cosmic neighbourhood.

Topics

- How the Solar System formed
- Formation of Earth
- Basic structure of the Solar System
- The eight planets
- Inner and outer planets
- Why Earth is suitable for life
- Introduction to Mars

Demonstrations

A scale/model representation of the Solar System will be used to introduce the planets.

Key question

> **“How did our home become the planet we live on today?”**

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6. SESSION 2 — HOW DO WE SEE SPACE?

Presenter 2 — Astronomy & Telescopes

Students will be introduced to the basic principles of astronomical observation.

Topics

- What is a telescope?
- Why telescopes are needed
- Refracting telescopes
- Reflecting telescopes
- Radio telescopes
- How telescopes collect light
- How astronomers observe distant objects
- Introduction to spectroscopy

Practical Demonstrations

A working telescope will be displayed and explained.

Where feasible, students will be allowed to observe a distant object through the telescope.

A simple spectroscope/diffraction-grating demonstration will be used to show how light can provide information about distant astronomical objects.(if possible)

Key question

> **"How can we learn about something that is millions or billions of kilometres away?"**

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7. SESSION 3 — CHOOSING OUR DESTINATION: WHY MARS?

Presenter 3 — Mission Planning & Planetary Science

Once students understand the Solar System, the programme introduces the mission objective.

Topics

- Why Mars?
- Mars compared with Earth
- Martian atmosphere

- Temperature
- Gravity
- Water ice
- Surface conditions
- Challenges of human exploration
- Previous Mars missions
- India’s Mars Orbiter Mission (Mangalyaan)

Interactive Activity

Students will examine a simplified Mars map and identify possible areas suitable for exploration based on factors such as terrain, sunlight and potential water resources.

The activity introduces the idea that spacecraft missions require careful planning before launch.

Key question

> **“If we were going to Mars, where would we explore and why?”**

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8. SESSION 4 — BUILDING OUR SPACECRAFT: THE ROCKET

Presenter 4 — Aerospace & Propulsion

The students will now move from mission planning to spacecraft design.

Topics

- Why rockets are needed
- Earth’s gravity
- Thrust
- Rocket engines
- Fuel
- Stages
- Payload
- Launch vehicles
- Basic trajectory concepts

Practical Activity

Practical Activity — Paper Rocket Flight Experiment

Students will be introduced to the basic principles of flight and rocket motion through a simple paper rocket experiment.

Each team will construct a basic paper rocket using easily available materials such as paper and tape. The rockets will then be launched.

The presenter will explain the basic effects of **initial force, gravity, air resistance and stability** on the rocket's motion.

Students will then modify selected design features such as the nose shape, fin configuration, body length or mass distribution and conduct a second flight test.

The activity will follow a simplified **design–test–observe–modify** cycle, introducing students to the basic principles of scientific experimentation and engineering design.

The activity will conclude by comparing the paper rocket with real launch vehicles and explaining how real rockets use engines to generate thrust and overcome Earth's gravity to place spacecraft into space.

Demonstration

A safe model rocket/water rocket or air-powered rocket demonstration will be conducted, subject to venue and safety approval.

Key question

> **"How do we make something leave Earth and travel into space?"**

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9. SESSION 5 — SURVIVING THE LANDING: MARS LANDER

Presenter 5 — Entry, Descent & Landing

Reaching Mars is only half the challenge.

The spacecraft must also survive its descent and land safely.

Topics

- Entry into a planetary atmosphere
- Descent
- Landing
- Importance of speed reduction
- Terrain hazards
- Landing site selection

- Why landing on Mars is difficult

Practical Demonstration

A physical Mars terrain model will be created using:

- Red surface material
- Craters
- Rocks
- Slopes
- Safe landing zones

A model lander will be used to demonstrate the landing challenge.

Students will observe how different landing conditions can affect the spacecraft.

Key question

“Can our spacecraft reach Mars without destroying itself when it lands?”

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10. SESSION 6 — EXPLORING MARS WITH A ROVER

Presenter 6 — Robotics & Exploration

After a successful landing, the mission begins its scientific exploration.

Topics

- What is a planetary rover?
- Why rovers are used
- Rover wheels and mobility
- Cameras
- Sensors
- Communication
- Autonomous navigation
- Scientific instruments
- Mars exploration missions

Practical Demonstration

A working rover model will be demonstrated on a small simulated Martian terrain.

The rover will be required to:

1. Navigate the terrain.

2. Detect/avoid obstacles.
3. Reach a designated scientific target.
4. Collect or identify a simulated sample.
5. Return to the lander.

Where feasible, an Arduino-based rover with sensors can be used to demonstrate how programming and robotics support space exploration.

Key question

> **"If humans cannot immediately go to Mars, how can we explore it?"**

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11. SESSION 7 — COMMUNICATING WITH EARTH

Presenter 7 — Space Communication & Mission Control

A Mars mission cannot succeed without communication.

Topics

- Spacecraft communication
- Radio signals
- Antennas
- Sending scientific data
- Receiving commands
- Communication delays
- Mission control
- Navigation and telemetry

Practical Demonstration

A simple classroom simulation will be used to demonstrate how information from a Mars rover can be communicated back to Earth without requiring any specialised communication equipment.

Selected students will act as the **Mars Rover** and **Mission Control**. The rover will be given a simple simulated scientific message, such as information about temperature, rock samples or the discovery of possible water/ice.

The message will be communicated using a simple predefined coding and decoding method. Mission Control will then decode the message and identify the information received from the rover.

The activity will introduce students to the basic communication process:

Information → Encode → Transmit → Receive → Decode

The presenter will then explain that real spacecraft use sophisticated communication systems and radio signals to exchange commands and scientific data with Earth.

The concept of **communication delay** will also be introduced through the simulation, demonstrating why communication with a spacecraft located millions of kilometres away cannot happen in the same way as a normal conversation.

Key question

> **"How does a spacecraft millions of kilometres away tell us what it has discovered?"**

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12. FINAL MISSION — MISSION COMPLETE

At the end of the individual sessions, all seven presenters will return to the central mission.

The students will be presented with a simulated mission scenario involving multiple failures.

For example:

- Communication problem
- Rover failure
- Power shortage
- Difficult terrain
- Scientific discovery
- Need to transmit data to Earth

Each presenter will briefly connect the problem to the scientific concept demonstrated during their session.

The programme will conclude with:

> **EARTH → LAUNCH → MARS → LAND → EXPLORE → COMMUNICATE → RETURN**

The final message will emphasize that space exploration is not the work of a single astronaut or scientist, but a collaboration involving engineers, programmers, astronomers, scientists, doctors, communication specialists and many other professionals.

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13. PROPOSED DAY SCHEDULE

Time	Programme
9:30 – 10:00	Registration & Team Formation
10:00 – 10:15	Inauguration & Mission Briefing
10:15 – 10:55	Session 1 — Earth & Solar System
10:55 – 11:35	Session 2 — Telescopes & Astronomy
11:35 – 11:50	Refreshment Break
11:50 – 12:30	Session 3 — Why Mars?
12:30 – 1:10	Session 4 — Rocket & Mission Design
1:10 – 2:00	Lunch Break
2:00 – 2:40	Session 5 — Mars Landing
2:40 – 3:20	Session 6 — Mars Rover & Robotics
3:20 – 4:00	Session 7 — Communication & Mission Control
4:00 – 4:15	Final Mission & Mission Complete
4:15 – 4:30	Certificates, Awards & Group Photograph

The schedule can be adjusted according to the availability of guests, inauguration arrangements and venue requirements.

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14. TEAM STRUCTURE

The approximately 100 students may be divided into **7 teams**.

Each team will be assigned a mission commander from the seven presenters.

Possible team names:

- Chandrayaan
- Mangalyaan
- Artemis
- Voyager
- Apollo
- Hubble
- Kepler

The presenters will act as team heads/facilitators rather than conventional lecturers.

This approach allows the presenters to guide their respective teams while ensuring that the entire audience remains engaged without requiring every student to participate in every activity.

Selected students can participate directly in demonstrations and practical activities, while the remaining students observe, discuss and make decisions with their teams.

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15. PROPOSED MODELS AND EQUIPMENT

The programme will make use of physical models and demonstrations wherever possible.

Space Models

- Solar System model
- Earth model
- Mars model
- Rocket model
- Mars lander model
- Mars rover model
- Satellite model

Astronomy Equipment

- Telescope
- Simple spectroscope/diffraction grating
- Astronomical images/videos

Practical Materials

- Cardboard
- Paper
- Foam board
- Plastic bottles
- Balloons
- Sand/flour
- Small balls
- Magnets
- LEDs
- Basic electronic components
- Arduino and sensors, if available

Audio-Visual Equipment

- Projector
- Screen
- Microphone
- Speakers
- Laptop

All demonstrations involving rockets, pressure, electronics or other potentially hazardous equipment will be conducted under appropriate supervision and safety procedures.

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16. EXPECTED LEARNING OUTCOMES

By the end of the programme, students should be able to:

- Explain the basic formation of the Solar System and Earth.
- Identify the planets and describe some of their key characteristics.
- Understand the basic purpose and working principles of telescopes.
- Explain why Mars is a major target for exploration.
- Understand the basic purpose of rockets and launch vehicles.
- Recognize the major stages of a Mars mission.
- Explain the role of landers and rovers.
- Understand how robotics assists planetary exploration.
- Recognize the importance of communication in space missions.
- Understand that space exploration requires multiple scientific and engineering disciplines.
- Develop curiosity and interest in science, technology and space exploration.

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17. INNOVATIVE ASPECT OF THE PROGRAMME

The primary innovation of this programme is that it does not treat astronomy as a sequence of independent lectures.

Instead, the entire programme is built around a **single continuous mission narrative**.

Students first learn about Earth and the Solar System because they need to understand where they are starting from.

They learn about telescopes because they need to understand how we observe distant worlds.

They study Mars because it becomes the mission destination.

They learn about rockets because the mission must leave Earth.

They learn about landing because reaching Mars is not enough.

They learn about rovers because the surface must be explored.

Finally, they learn about communication because the discoveries must reach scientists on Earth.

Thus, every session provides the scientific foundation for the next stage of the mission.

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18. PROPOSED IMPACT

The programme is expected to provide students with a learning experience that is:

- Interactive
- Demonstration-based
- Practical
- Team-oriented
- Scientifically grounded
- Age-appropriate
- Visually engaging
- Connected to real-world space missions

Rather than simply remembering facts about planets and spacecraft, students will experience how different scientific concepts come together to accomplish a space mission.

The programme also aims to encourage students to consider future careers in:

- Astronomy
- Aerospace engineering
- Robotics
- Computer science
- Electronics
- Physics
- Earth and planetary science
- Data science
- Space technology

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19. CONCLUSION

Mission Mars: From Earth to the Red Planet is proposed as an immersive academic outreach programme designed to introduce school students to the fascinating world of space science through a combination of storytelling, demonstrations, physical models, technology and practical activities.

The programme transforms the traditional approach of “learning about space” into an experience of **planning and completing a space mission**.

Beginning with the formation of Earth and ending with the exploration of Mars, the programme will demonstrate how astronomy, physics, engineering, robotics, computer science and communication technology work together in modern space exploration.

Through this initiative, we hope to encourage students to look beyond the textbook, develop scientific curiosity, and see themselves as potential contributors to the future of space exploration.

> **“Every great space mission begins with a question.**
> **This programme begins by asking students:**
> **What if we could go to Mars?”**

Proposed Programme Title

🚀 MISSION MARS

From Earth to the Red Planet

An Interactive Space Science Outreach Programme for School Students

Proposed in collaboration with Pondicherry Science Forum